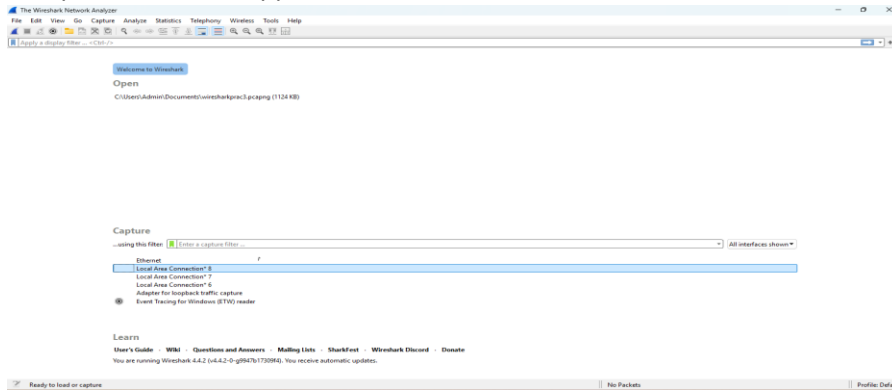


## PRACTICAL 3

Aim : Capturing and analyzing network packets using Wireshark

Steps:

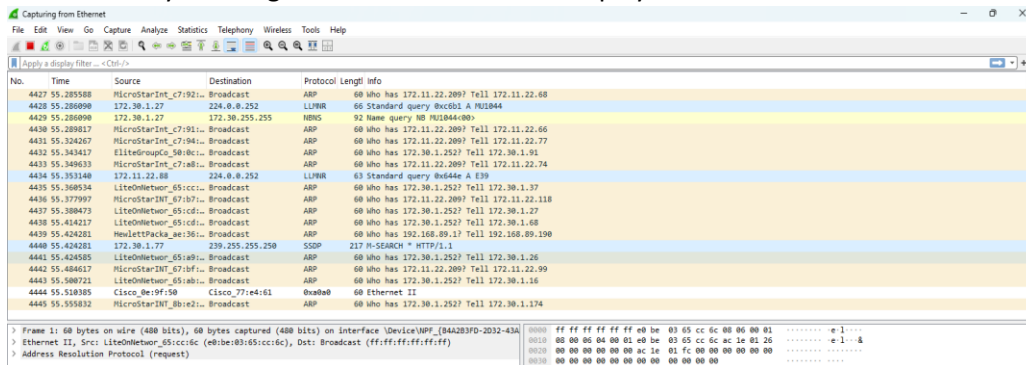
1. Download WireShark:
2. Setup and start the App:



3. Double click on Preferred Internet Connection:  
In this Case for us it will be Ethernet:



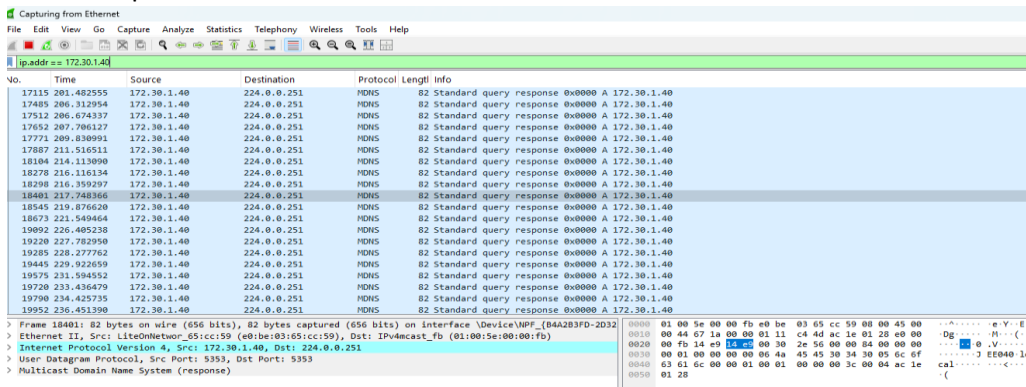
4. All the activity on the given connection will be Displayed:



5. You can use different Filter Commands to filter out the activity:

- Display packets based on specific IP-address

`ip.addr == 172.30.1.40`



- Display packets which are coming from specific IP-address

**ip.src == 172.11.22.67**

| Capturing from Ethernet                                                                                                                                                                                                                                                                                                                                                                                          |            |              |                |          |        |                                                  |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------|--------------|----------------|----------|--------|--------------------------------------------------|
| File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help                                                                                                                                                                                                                                                                                                                                       |            |              |                |          |        |                                                  |
| ip.src == 172.11.22.67                                                                                                                                                                                                                                                                                                                                                                                           |            |              |                |          |        |                                                  |
| No.                                                                                                                                                                                                                                                                                                                                                                                                              | Time       | Source       | Destination    | Protocol | Length | Info                                             |
| 15832                                                                                                                                                                                                                                                                                                                                                                                                            | 187.039175 | 172.11.22.67 | 172.11.255.255 | NBNS     | 92     | Name query NB E39<00>                            |
| 15868                                                                                                                                                                                                                                                                                                                                                                                                            | 187.289288 | 172.11.22.67 | 224.0.0.251    | MDNS     | 69     | Standard query 0x0000 A E39.local, "QM" question |
| 15908                                                                                                                                                                                                                                                                                                                                                                                                            | 187.895208 | 172.11.22.67 | 172.11.255.255 | NBNS     | 92     | Name query NB E39<00>                            |
| 18280                                                                                                                                                                                                                                                                                                                                                                                                            | 216.128695 | 172.11.22.67 | 172.11.255.255 | NBNS     | 92     | Name query NB E39<00>                            |
| 18281                                                                                                                                                                                                                                                                                                                                                                                                            | 216.128878 | 172.11.22.67 | 224.0.0.251    | MDNS     | 69     | Standard query 0x0000 A E39.local, "QM" question |
| 18282                                                                                                                                                                                                                                                                                                                                                                                                            | 216.129141 | 172.11.22.67 | 224.0.0.252    | LLMNR    | 63     | Standard query 0x748d A E39                      |
| 18305                                                                                                                                                                                                                                                                                                                                                                                                            | 216.549304 | 172.11.22.67 | 224.0.0.252    | LLMNR    | 63     | Standard query 0x748d A E39                      |
| 18336                                                                                                                                                                                                                                                                                                                                                                                                            | 216.892693 | 172.11.22.67 | 172.11.255.255 | NBNS     | 92     | Name query NB E39<00>                            |
| 18353                                                                                                                                                                                                                                                                                                                                                                                                            | 217.142797 | 172.11.22.67 | 224.0.0.251    | MDNS     | 69     | Standard query 0x0000 A E39.local, "QM" question |
| 18389                                                                                                                                                                                                                                                                                                                                                                                                            | 217.658421 | 172.11.22.67 | 172.11.255.255 | NBNS     | 92     | Name query NB E39<00>                            |
| 21291                                                                                                                                                                                                                                                                                                                                                                                                            | 249.509159 | 172.11.22.67 | 172.11.255.255 | NBNS     | 92     | Name query NB E39<00>                            |
| 21292                                                                                                                                                                                                                                                                                                                                                                                                            | 249.509324 | 172.11.22.67 | 224.0.0.251    | MDNS     | 69     | Standard query 0x0000 A E39.local, "QM" question |
| 21293                                                                                                                                                                                                                                                                                                                                                                                                            | 249.509996 | 172.11.22.67 | 224.0.0.252    | LLMNR    | 63     | Standard query 0xbdb1 A E39                      |
| 21329                                                                                                                                                                                                                                                                                                                                                                                                            | 249.929686 | 172.11.22.67 | 224.0.0.252    | LLMNR    | 63     | Standard query 0xbdb1 A E39                      |
| 21358                                                                                                                                                                                                                                                                                                                                                                                                            | 250.272731 | 172.11.22.67 | 172.11.255.255 | NBNS     | 92     | Name query NB E39<00>                            |
| 21375                                                                                                                                                                                                                                                                                                                                                                                                            | 250.522729 | 172.11.22.67 | 224.0.0.251    | MDNS     | 69     | Standard query 0x0000 A E39.local, "QM" question |
| 21429                                                                                                                                                                                                                                                                                                                                                                                                            | 251.038977 | 172.11.22.67 | 172.11.255.255 | NBNS     | 92     | Name query NB E39<00>                            |
| 24246                                                                                                                                                                                                                                                                                                                                                                                                            | 282.878131 | 172.11.22.67 | 172.11.255.255 | NBNS     | 92     | Name query NB E39<00>                            |
| 24247                                                                                                                                                                                                                                                                                                                                                                                                            | 282.878131 | 172.11.22.67 | 224.0.0.251    | MDNS     | 69     | Standard query 0x0000 A E39.local, "QM" question |
| 24248                                                                                                                                                                                                                                                                                                                                                                                                            | 282.878575 | 172.11.22.67 | 224.0.0.252    | LLMNR    | 63     | Standard query 0xd068 A E39                      |
| > Frame 18389: 92 bytes on wire (736 bits), 92 bytes captured (736 bits) on interface \Device\NPF_{B4A2B3FD-2D32-43...}                                                                                                                                                                                                                                                                                          |            |              |                |          |        |                                                  |
| > Ethernet II, Src: MicroStarInt_c7:95:4d (d4:3d:7e:c7:95:4d), Dst: Broadcast (ff:ff:ff:ff:ff:ff)                                                                                                                                                                                                                                                                                                                |            |              |                |          |        |                                                  |
| > Internet Protocol Version 4, Src: 172.11.22.67, Dst: 172.11.255.255                                                                                                                                                                                                                                                                                                                                            |            |              |                |          |        |                                                  |
| > User Datagram Protocol, Src Port: 137, Dst Port: 137                                                                                                                                                                                                                                                                                                                                                           |            |              |                |          |        |                                                  |
| > NetBIOS Name Service                                                                                                                                                                                                                                                                                                                                                                                           |            |              |                |          |        |                                                  |
| 0000 ff ff ff ff ff ff d4 3d 7e c7 95 4d 08 00 45 00 .....M..E<br>0010 00 4e 9e 44 00 00 80 11 2e 01 ac 0b 16 43 ac 0b .....N.D.....C<br>0020 ff ff 00 89 00 89 00 3a 82 f8 c6 ad 01 10 00 01 .....E PDDJ3CAC<br>0030 00 00 00 00 00 00 20 45 46 44 44 4a 43 41 43 .....E PDDJ3CAC<br>0040 41 43 41 43 41 43 41 43 41 43 41 43 41 43 41 43 ACACACAC ACACACAC<br>0050 41 43 41 43 41 41 41 00 00 20 00 01 ACACAAA |            |              |                |          |        |                                                  |

- Display packets which are having specific IP-address destination

**ip.dst == 192.30.255.255**

| Capturing from Ethernet                                                                                                                                                                                                                                                                                                                                                                                    |          |              |                |          |        |                               |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|--------------|----------------|----------|--------|-------------------------------|
| File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help                                                                                                                                                                                                                                                                                                                                 |          |              |                |          |        |                               |
| ip.dst == 192.30.255.255                                                                                                                                                                                                                                                                                                                                                                                   |          |              |                |          |        |                               |
| No.                                                                                                                                                                                                                                                                                                                                                                                                        | Time     | Source       | Destination    | Protocol | Length | Info                          |
| 20                                                                                                                                                                                                                                                                                                                                                                                                         | 0.282242 | 172.30.1.27  | 172.30.255.255 | NBNS     | 92     | Name query NB PU1043<00>      |
| 30                                                                                                                                                                                                                                                                                                                                                                                                         | 0.395436 | 172.30.1.174 | 172.30.255.255 | NBNS     | 110    | Registration NB PU1174<00>    |
| 31                                                                                                                                                                                                                                                                                                                                                                                                         | 0.395436 | 172.30.1.174 | 172.30.255.255 | NBNS     | 110    | Registration NB WORKGROUP<00> |
| 32                                                                                                                                                                                                                                                                                                                                                                                                         | 0.395436 | 172.30.1.174 | 172.30.255.255 | NBNS     | 92     | Name query NB JEG040<00>      |
| 89                                                                                                                                                                                                                                                                                                                                                                                                         | 1.040789 | 172.30.1.27  | 172.30.255.255 | NBNS     | 92     | Name query NB PU1043<00>      |
| 93                                                                                                                                                                                                                                                                                                                                                                                                         | 1.160989 | 172.30.1.174 | 172.30.255.255 | NBNS     | 110    | Registration NB PU1174<00>    |
| 94                                                                                                                                                                                                                                                                                                                                                                                                         | 1.160989 | 172.30.1.174 | 172.30.255.255 | NBNS     | 110    | Registration NB WORKGROUP<00> |
| 95                                                                                                                                                                                                                                                                                                                                                                                                         | 1.160989 | 172.30.1.174 | 172.30.255.255 | NBNS     | 92     | Name query NB JEG040<00>      |
| 140                                                                                                                                                                                                                                                                                                                                                                                                        | 1.800586 | 172.30.1.27  | 172.30.255.255 | NBNS     | 92     | Name query NB PU1043<00>      |
| 196                                                                                                                                                                                                                                                                                                                                                                                                        | 1.925447 | 172.30.1.174 | 172.30.255.255 | NBNS     | 110    | Registration NB WORKGROUP<00> |
| 197                                                                                                                                                                                                                                                                                                                                                                                                        | 1.925447 | 172.30.1.174 | 172.30.255.255 | NBNS     | 110    | Registration NB PU1174<00>    |
| 374                                                                                                                                                                                                                                                                                                                                                                                                        | 3.024540 | 172.30.1.27  | 172.30.255.255 | NBNS     | 92     | Name query NB PU1037<1c>      |
| 423                                                                                                                                                                                                                                                                                                                                                                                                        | 4.281645 | 172.30.1.27  | 172.30.255.255 | NBNS     | 92     | Name query NB PU1006<00>      |
| 438                                                                                                                                                                                                                                                                                                                                                                                                        | 4.377126 | 172.30.1.37  | 172.30.255.255 | NBNS     | 92     | Name query NB PU1037<1c>      |
| 514                                                                                                                                                                                                                                                                                                                                                                                                        | 5.042716 | 172.30.1.27  | 172.30.255.255 | NBNS     | 92     | Name query NB PU1006<00>      |
| 519                                                                                                                                                                                                                                                                                                                                                                                                        | 5.130808 | 172.30.1.37  | 172.30.255.255 | NBNS     | 92     | Name query NB PU1037<1c>      |
| 566                                                                                                                                                                                                                                                                                                                                                                                                        | 5.883613 | 172.30.1.27  | 172.30.255.255 | NBNS     | 92     | Name query NB PU1006<00>      |
| 658                                                                                                                                                                                                                                                                                                                                                                                                        | 6.501610 | 172.30.1.49  | 172.30.255.255 | NBNS     | 92     | Name query NB WORKGROUP<1c>   |
| 714                                                                                                                                                                                                                                                                                                                                                                                                        | 7.274284 | 172.30.1.49  | 172.30.255.255 | NBNS     | 92     | Name query NB WORKGROUP<1c>   |
| > Frame 20: 92 bytes on wire (736 bits), 92 bytes captured (736 bits) on interface \Device\NPF_{B4A2B3FD-2D32-43...}                                                                                                                                                                                                                                                                                       |          |              |                |          |        |                               |
| > Ethernet II, Src: LiteOnNetwor_65:cd:d19 (e0:be:03:65:cd:19), Dst: Broadcast (ff:ff:ff:ff:ff:ff)                                                                                                                                                                                                                                                                                                         |          |              |                |          |        |                               |
| > Internet Protocol Version 4, Src: 172.30.1.27, Dst: 172.30.255.255                                                                                                                                                                                                                                                                                                                                       |          |              |                |          |        |                               |
| > User Datagram Protocol, Src Port: 137, Dst Port: 137                                                                                                                                                                                                                                                                                                                                                     |          |              |                |          |        |                               |
| > NetBIOS Name Service                                                                                                                                                                                                                                                                                                                                                                                     |          |              |                |          |        |                               |
| 0000 ff ff ff ff ff ff e0 be 03 65 cd 19 08 00 45 00 .....e...E<br>0010 00 4e 4d d6 00 00 80 11 93 71 ac 1e 01 1b ac 1e .....q.....<br>0020 ff ff 00 89 00 89 00 3a 81 27 04 7b 01 10 00 01 .....E.....<br>0030 00 00 00 00 00 00 20 45 46 46 44 42 44 41 44 .....E NFFD0DAD<br>0040 45 44 44 43 41 43 41 43 41 43 41 43 41 43 41 43 EDDCACAC ACACACAC<br>0050 41 43 41 43 41 41 41 00 00 20 00 01 ACACAAA |          |              |                |          |        |                               |

- Display packets which are using http protocol

**http**

| No.   | Time       | Source          | Destination     | Protocol | Length | Info                                       |
|-------|------------|-----------------|-----------------|----------|--------|--------------------------------------------|
| 58    | 74.985092  | 192.168.1.4     | 192.168.1.1     | HTTP     | 250    | GET /rootDesc.xml HTTP/1.1                 |
| 62    | 74.987756  | 192.168.1.1     | 192.168.1.4     | HTTP/X.. | 1234   | HTTP/1.1 200 OK                            |
| 972   | 129.457310 | 192.168.1.4     | 172.217.166.174 | HTTP     | 1000   | GET / HTTP/1.1                             |
| 975   | 129.542230 | 172.217.166.174 | 192.168.1.4     | HTTP     | 594    | HTTP/1.1 301 Moved Permanently (text/html) |
| 39156 | 277.292187 | 192.168.1.4     | 117.18.237.29   | OCSP     | 137    | Request                                    |
| 39157 | 277.314544 | 117.18.237.29   | 192.168.1.4     | OCSP     | 842    | Response                                   |
| 39168 | 277.419340 | 192.168.1.4     | 117.18.237.29   | OCSP     | 137    | Request                                    |
| 39169 | 277.463638 | 117.18.237.29   | 192.168.1.4     | OCSP     | 842    | Response                                   |
| 39204 | 279.409683 | 192.168.1.4     | 23.57.219.27    | OCSP     | 137    | Request                                    |
| 39206 | 279.420870 | 23.57.219.27    | 192.168.1.4     | OCSP     | 712    | Response                                   |

- Display packets which are using http request

**http.request**

| No. | Time      | Source      | Destination     | Protocol | Length | Info                       |
|-----|-----------|-------------|-----------------|----------|--------|----------------------------|
| 40  | 50.307358 | 192.168.1.3 | 239.255.255.250 | SSDP     | 167    | M-SEARCH * HTTP/1.1        |
| 41  | 50.607228 | 192.168.1.3 | 239.255.255.250 | SSDP     | 167    | M-SEARCH * HTTP/1.1        |
| 46  | 60.015835 | 192.168.1.3 | 239.255.255.250 | SSDP     | 167    | M-SEARCH * HTTP/1.1        |
| 47  | 60.306194 | 192.168.1.3 | 239.255.255.250 | SSDP     | 167    | M-SEARCH * HTTP/1.1        |
| 48  | 60.605851 | 192.168.1.3 | 239.255.255.250 | SSDP     | 167    | M-SEARCH * HTTP/1.1        |
| 49  | 70.031605 | 192.168.1.3 | 239.255.255.250 | SSDP     | 167    | M-SEARCH * HTTP/1.1        |
| 50  | 70.321279 | 192.168.1.3 | 239.255.255.250 | SSDP     | 167    | M-SEARCH * HTTP/1.1        |
| 51  | 70.626289 | 192.168.1.3 | 239.255.255.250 | SSDP     | 167    | M-SEARCH * HTTP/1.1        |
| 53  | 73.874454 | 192.168.1.4 | 239.255.255.250 | SSDP     | 175    | M-SEARCH * HTTP/1.1        |
| 58  | 74.985092 | 192.168.1.4 | 192.168.1.1     | HTTP     | 250    | GET /rootDesc.xml HTTP/1.1 |
| 62  | 74.987756 | 192.168.1.1 | 192.168.1.4     | HTTP     | 1234   | HTTP/1.1 200 OK            |

